Jashore University of Science and Technology

Bachelor of Science Electrical and Electronic Engineering

1st semester of 1st year (2025–2026)

Course no.: 053311 PHY 1101 **Course title: Physics**

Class test no.: 01 **Date: 29 August 2026**

Roll:	

1. Is the speed of light in vacuum the same for all inertial observers, regardless of their relative motion? [1]

Yes No

2. Does Galilean transformation remain valid for describing the motion of objects moving at speeds comparable to the speed of light? [1]

Yes No

3. Does a moving clock run slower relative to a stationary observer? [1]

Yes No

4. Is the rest mass of an object changed merely because it is observed from a different inertial reference frame? [1]

Yes No

5. If two events are simultaneous in one inertial frame, must they also be simultaneous in every other inertial frame? [1]

Yes No

6. A particle has a rest energy of mc^2. Calculate the total relativistic energy of a particle moving with a speed of $0.60c$ in terms of its rest energy. [2.5]

7. A spacecraft travels at $0.60c$ relative to Earth. If 10 minutes elapse on a clock inside the spacecraft, how much time is measured by an observer on Earth? [2.5]